EXAM NUMBER: .....

KMTC/QP-08/TIS

Kenya Medical Training College

Department of Medical Laboratory Sciences

Final Qualifying Examinations 2017

Diploma in Medical Laboratory Sciences (DMLS)

Paper: 05-A Immunology (100 marks)

Date:	Day:	Time:
21/06/2017	Wednesday	8.30 - 10.30 am

Instructions

1. Mobile phones are not allowed in the examination room
2. Write your exam number on the answer book used.
3. Do not write your name on the answer book.
4. Attempt both sections, I and II.

Section I comprises of M.C.Q.

5. Write MCQ answers in capitals on the first page of the answer book.
6. Be careful, as $^1/_2$ mark will be deducted for every wrong response.

Section II comprises of Essay Questions.

7. Attempt all the questions. Make sure you understand the question before answering.

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SECTION I: (MCQ 40 MARKS)

1) The following statements describe variolation except:-
   a) It was the first method used to induce immunity
   b) Was invented by Chinese medicine men to confer immunity against small pox
   c) it is currently being practiced in developed world to induce immunity
   d) Was the first attempt to induce active immunity
   e) Involved inoculation of purified microorganisms to induce immunity

2) Professional antigen presenting cells is/are:-
   a) Neutrophils
   b) Basophils
   c) Eosinophils
   d) Macrophages
   e) T-cells

3) Which of the following is **not** an effector mechanism of antibody binding *in-vivo*?
   a) Apoptosis
   b) Neutralization
   c) Lysis
   d) Phagocytosis
   e) Agglutination

4) Deficiency of complement proteins or their regulators may result in:-
   a) Immune complex disease
   b) Increased susceptibility to infections
   c) Gout
   d) Kidney disease
   e) Increased red cell lysis

5) Which of the following is protective chemical in tears and saliva?
   a) Complement
   b) Lysozyme
   c) Mucus
   d) Interferons
   e) Sebum

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6) Which of the following is not an endocrine autoimmune disease?
   a) Type I diabetes mellitus
   b) Autoimmune pancreatitis
   c) Hashimotos thyroiditis
   d) Addison's disease
   e) Psoriasis

7) Cytotoxic T-lymphocytes:-
   a) Have CD_4 receptors
   b) Recognize antigen presented by antigen presenting cells
   c) Recognize antigen peptides in association with MHC-1 molecules
   d) Kill target cells by inducing apoptosis.
   e) Stimulate B-lymphocytes to produce antibodies

8) Which of the following is/are part of the major histocompatibility complex class II:-
   a) HLA-DR
   b) HLA-DP
   c) HLA-A
   d) HLA-B
   e) HLA-C

9) Which of the following is **not** type II hypersensitivity reaction?
   a) Haemolytic transfusion reaction
   b) Myasthenia gravis
   c) Good pastures syndrome
   d) Haemolytic disease of the foetus and newborn
   e) Rheumatoid arthritis

10) Transient hypogammaglobulinemia of infancy:-
   a) Occurs at 3 months of age
   b) Is due to placental absorption of IgG.
   c) Is more pronounced in premature babies
   d) Is linked to autoimmune disease in later life
   e) Involves only the IgA

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11) Which of the following is the primary defect in chronic granulomatous dise

a) Neutrophil production in the bone marrow
b) Neutrophil chemotaxis
c) Opsonization
d) Neutrophil intracellular killing of organisms
e) Cytotoxic T-cell activity

12) A reaction between antibody and soluble antigens is demonstrated by:-

a) Agglutination test
b) Absorption test
c) Haemagglutination
d) Coomb's test
e) Precipitation test

13) The alternate complement pathway:-

a) Is initiated by the formation of antigen antibody reactions classical pathway
b) Is activated by factors such as endotoxin
c) Is activated by haematopoietic cells
d) Is predominantly a non-antibody initiated pathway.
e) Is a potent mediator of apoptosis

14) Lymphokines:-

a) Stimulate increased antibody synthesis
b) Do not induce immunoglobulin class switching
c) Induce maturation of immature cytotoxic T-cells
d) Do not induce maturation of immature T-cells
e) Are not classified

15) Among the following lymphoid tissues select one that belongs to the secondary lymphoid tissue:-

a) Yolk Sac
b) Foetal liver
c) Lymph nodes
d) Bone marrow } Primary lymphoid tissue
e) Thymus

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Strongly immunogenic tumours appear:-

a) In virtually all cancers
b) Only in lymphoma and leukemia
c) Only in experimental animals
d) Only in elderly patients
e) In immunosuppressed patients

17) Negative feedback on adaptive B-cell responses is mediated by:-

a) Antigen specific IgM
b) FC gamma receptors on macrophages
c) Antigen specific IgG
d) Antigen neutralization
e) $F(ab)_2$ anti-mu

18) Western blots are particularly used to detect:-

a) Carbohydrates
b) Proteins e.g. the P_{24} in HIV.
c) Lipid
d) RNA
e) DNA

19) Components of acquired immunity against extra cellular infections include:-

a) Antibody
b) Cytokine
c) Lymphokine
d) Complement
e) T-lymphocyte

20) Which of these cells are involved in the fight against tumor development?

a) B-cells
b) Platelets
c) Cytotoxic lymphocytes
d) Natural killer cells
e) Normoblasts

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21) Rejection of a transplanted allograft may be avoided by:-

a) Gender matching
b) Matching for CMV status
c) HLA matching
d) Parental matching
e) Immunosuppression

22) The spleen is involved in:-

a) Antigen dependent maturation of normoblasts
b) Filtration of antigens in blood
c) Antibody production
d) Regulation of hormonal immunity
e) Maturation of B-lymphocytes

23) In radioimmunoassay, the antigen is labeled with the following:-

a) Iodine
b) Substrate
c) Phosphatase
d) Enzyme
e) Methylene blue

24) An adjuvant in vaccine production:-

a) Prevents pain development in the recipient
b) Makes the antigen readily degradable
c) Prevents back mutation of attenuated microbes
d) Enhances immunogenicity of the vaccine
e) Attenuates the microbes

25) If an individual is genetically unable to make "J" chains, which immunoglobulins would be affected?

a) sIgA
b) IgA
c) IgM
d) IgG
e) IgD

26) A presenting cells that activate helper T-Cells must express the following on surface:

a) MHC - Class III
b) MHC - Class I
c) FcER
d) IgE
e) MHC - Class II

27) Delayed hypersensitivity reaction is characterized by:-

a) Edema without cellular infiltrate
b) Infiltrate composed of neutrophils
c) Infiltrate of T helper cells and macrophages
d) Infiltrate of eosinophils
e) An infiltrate of basophils

28) Central tolerance takes place in:-

a) Lymph nodes
b) Thymus
c) MALT
d) Liver
e) Lungs

29) Select the one that can neutralize toxins:-

a) Complement
b) Toxoid
c) PGE_2
d) Proteolytic enzymes
e) Antibody

30) Neutrophil chemotaxis is mediated by:-

a) C_5b
b) E-selectin
c) IL-8
d) CCL_2
e) C_3a

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31) T-lymphocyte selection occurs in the:-

a) Thymus
b) Bone marrow – B-lymphocyte
c) Lymph nodes
d) Spleen
e) Thyroid

SECTION II (ESSAY 60 MARKS)

1) Describe type II hypersensitivity.
(20 marks)

2)
a) The study of Major Histocompatibility Complex (MHC) in immunology is important, justify. (10 marks)
b) Outline the development of cell mediated immunity in man.
(10 marks)

3)
a) Differentiate between innate and adaptive immunity.
b) State mechanisms by which tumours escape the immune system.
c) List five (5) advantages of toxoid vaccines.
d) List five (5) examples of acute phase proteins.
(20 marks)

2 Cover page
2 papers the same

Sep 2016
March 2017

Immunology marking keys
July 2017

MCQ

1. C, E
2. D
3. A
4. B
5. B
6. E
7. C, D
8. A, B
9. C, E
10. C
11. D
12. E
13. B, D
14. A, C
15. C
16. E
17. C
18. B
19. A, D
20. C, D
21. E, C
22. B
23. A
24. D
25. A, C, D
26. E
27. C
28. B, D
29. E
30. C
31. A

Essay marking key (60 marks)

1. HYPERSENSITIVITY TYPE II

Definition - Hypersensitivity type II reaction (Cytotoxic hypersensitivity) Is an antibody mediated immune response directed against cell membrane antigens that affect a variety of organs and tissues. (2 marks)

Causes/Aetiologies

Intrinsic antigens – naturally occurring antigens (in this case, we would term the reaction "autoimmune")

Extrinsic antigens - exogenous substance such as a drug metabolite that has absorbed to a cell surface or exogenous chemicals (Haptens)

Any aetiology mentioned 2 marks

Mechanism

- ✓ Generation of humoral immune response and sensitization of cells immunity where IgG antibodies are absorbed on corresponding cell membrane antigen marks)
- ✓ Destruction of cells (6 marks)
  - Antibody –mediated destruction of cells
    - Opsonization and phagocytosis
    - C3b & C5b opsonizes antigens
    - Plasma cells produce IgG opsonizes antigens
    - Ingested by phagocytes
  - Antibody mediated complement activation
    - Acute inflammation: C3a and C5a are chemotaxis factor to neutrophil and monocytes
    - Cytolysis through the formation of MAC

Examples

- ✓ Non-autoimmune type II hypersensitivity
  - Blood transfusion reaction (immediate & delayed trx)
  - Haemolytic disease of the newborn
- ✓ Auto-immune type II hypersensitivity
  - Autoimmune haemolytic anaemia
  - Drug-induced haemolytic anaemia (DIHA)
  - Granulocytopenia
  - Thrombocytopenia purpura

Any 2 examples mentioned 2 marks each

Diagnosis (2marks)

- ✓ Antibody screening

2 a)

1. Role in organ transplants (2 marks)

Transplantation of healthy organs to replace deceased or damaged one demands the donor and recipient should be approximately close in terms of HLA genes. If t not considered there is great risk of the graft being rejected.

2. Paternity cases/disputes (2 marks)

Conclusive evidence of a child belonging to a family or not is made on the basis o antigens expressed by the child. If these are nearly similar to both parents then tl child's belongs but if little resemblance, then the child is not.

3. Regulation of immunity to intracellular infections (2 marks)
Activation of cellular immunity relies on the ability of T-lymphocytes to recognize peptides - MHC complexes. Absence of peptides MHC complexes results in no activation of T-Cells and hence no cellular immunity also T-dependent B-cell activation will not occur.

4. Forensic application (2 marks)
Where it is not possible to identify a body by physical characteristics the option usually employed in DNA typing of the body and purported relatives.

5. Application to transfusion (2 marks)
Non-red cell transfusion reactions have been attributed to leucocyte antigens. A common example is graft versus host diseases.

6. Academic purposes
a) Establishment of family trees
b) Tracing population migration

2b)
- Cell mediated immunity is dependent on the sub population of T-lymphocytes and macrophages and ~~their pro~~ducts. (1 mark)
- The macrophages present antigen via their surface MHC to T-lymphocytes. (1 mark)
- The T-lymphocytes recognize antigen through their specific T - cell receptors (TCRs) (1 mark)
- A specific T - cell is activated and differentiates into clone. (1 mark)
- Activated T-Lymphocytes develop to effector and memory T (1 mark)
- T-effector cell synthesize and secrete cytokines. (1 mark)
- The secreted cytokines stimulate other CMI and those of the humoral immune response these include:- (1 mark)
- Macrophages, monocytes and lymphocytes are attracted to the site of infection (1 mark)
- Macrophages directly kill intracellular pathogens (1 mark)
- $CD8^+$ lymphocytes are activated to kill virus and infected cells and tumour cells and initiate graft rejection. (1 mark)
- B - Cells are stimulated to differentiate into plasma cells and memory B. (1 mark)
- NK activity is enhanced. (1 mark)

a) Differentiate between innate and adaptive immunity

INNATE	ADAPTIVE
Natural	Acquired
Less diverse	Highly diverse
Uses PRR and PAMP receptors	Uses BCR and TCR
Antigen independent	Antigen dependent
Lacks memory	Has memory
Acts immediately	Required sometime to act

1 mark for every correct difference

b) State mechanisms by which tumours escape the immune system

i. Loss of antigen expression
ii. Production of anti-tumor responses
iii. Production of myeloid derived suppressor cells (MDSCs) which induce IL-10 whic inhibits inflammatory function.
iv. Production of T-regulatory cells which suppress T-cell responses.
v. Production of tumour associated macrophages which suppress T-cells.

1 mark for every correct mechanism

Q3c. State the advantage of toxoid vaccines

1 Prevention of certain bacterial infections
2 Stimulate antibody mediated immunity
3 They do not revert to virulence
4 Fairly stable over long period/conditions
5 Adaptable to DNA methods of production

1 mark for every correct advatage

Q3d. Acute phase proteins consist of a variety of serum proteins. State five examples of acute phase proteins

i. C-reactive protein
ii. Serum amyloid
iii. Fibrinogen
iv. Ceruloplasmin
v. Haptoglobin
vi. Mannose binding protein
vii. Alpha -1 antitrypsin

1 mark for every correct response